PARKINSON'S DISEASE DETECTION

Reliability-Aware Multi-Modal Analysis System

Gait · Voice · Uncertainty Quantification · Conformal Prediction · Cross-Modal Fusion

Master Technical Project Document

CSE AI / ML Final Year Project | 2025–2026

■ IMPORTANT DISCLAIMER

*This system is a **research prototype** developed as a final-year academic project. It is **NOT** validated for clinical use and must **NOT** be used for medical diagnosis, treatment, or any clinical decision-making. All experimental results are strictly exploratory and were obtained on small-scale publicly available research datasets.*

1. Executive Summary

This document presents the complete technical record of a final-year CSE AI/ML project that builds a **reliability-aware, multi-modal Parkinson's disease detection system**. The system does not merely output a binary prediction — it quantifies how much the prediction can be trusted, using a four-layer evidence stack: ensemble agreement, Bayesian uncertainty, conformal coverage, and cross-modal consistency.

Project Mission

Input: Gait sensor data (vertical ground-reaction force) or Speech recordings (spontaneous dialogue / read text).

Output: Prediction (Parkinson's Disease / Healthy Control) + Reliability Decision (ACCEPT / FLAG).

Core Question Answered: "Can the model's own prediction be trusted, or should a clinician be consulted?"

All experimental results are frozen. No models are retrained for this document.

179	69 + 110	5 Phases	M=5 + N=30
Tests Passing	Voice + Gait Tests	Uncertainty Layers	Ensemble + MC Passes

Both pipelines were developed across ten successive engineering phases (Phases 1–10) and validated against held-out test subjects. The Streamlit demo application unifies all results into a live, interactive panel demonstration.

2. Introduction and Motivation

1.1 The Clinical Problem

Parkinson's Disease (PD) is a progressive neurodegenerative disorder affecting approximately 8.5 million people worldwide (WHO, 2023). Characteristic motor symptoms — tremor, rigidity, bradykinesia, and postural instability — manifest in measurable changes to gait patterns and vocal tract dynamics long before formal clinical diagnosis. Early detection is clinically important: evidence suggests that neuroprotective interventions may be more effective when applied at early disease stages.

Conventional diagnostic approaches rely on clinician observation, UPDRS scoring, and expensive neuroimaging. Machine learning systems trained on objective sensor data offer an opportunity to automate screening, but only if the system can honestly signal when its predictions are not trustworthy.

1.2 Limitations of Conventional ML Systems

- **Overconfident softmax outputs**: Standard neural networks output probabilities that are systematically over-confident, especially under distribution shift.
- **Single-modality fragility**: Systems relying on one sensor type (gait OR voice) may fail when that modality is unreliable (sensor dropout, acoustic noise, sensor placement).
- **No abstention mechanism**: Most systems always produce a hard prediction, even when the evidence is ambiguous.
- **No cross-modal consistency check**: Independent modality predictions may directly contradict each other without any flag being raised.
- **No formal coverage guarantees**: Conformal prediction, which provides finite-sample marginal coverage, is rarely implemented in clinical ML prototypes.

1.3 Project Contribution

This project addresses each limitation above by engineering a system with **five reliability layers**:

Layer	Method	What It Detects
1 — Calibration	Temperature Scaling (post-hoc)	Overconfident probability estimates
2 — Bayesian UQ	MC Dropout (N=30 stochastic passes)	Epistemic uncertainty, model disagreement
3 — Ensemble	Deep Ensemble (M=5 seeds)	Prediction variance across independent models
4 — Set Coverage	Split Conformal Prediction (α=0.1)	Formal 90% marginal coverage guarantee
5 — Fusion	Cross-Modal Reliability Fusion	Gait ↔ Voice evidence conflict detection

Table 1. *Five reliability layers implemented in this project.*

3. Datasets

2.1 Gait Dataset — PhysioNet Gait-in-Parkinson's Disease

The gait analysis pipeline uses the publicly available **PhysioNet Gait-in-Parkinson's Disease 1.0.0** dataset (Hausdorff et al., 2000). It contains vertical ground-reaction force (VGRF) recordings from 16 force-sensitive resistors (8 per foot) sampled at **100 Hz**.

Property	Value
Dataset Source	PhysioNet (physionet.org/content/gaitndd)
Sampling Rate	100 Hz
Sensor Channels	16 (8 left foot + 8 right foot FSRs)
Cohort Ga (Galveston)	47 subjects (PD + HC)
Cohort Ju (Jülich)	54 subjects (PD + HC)
Cohort Si (Siegburg)	64 subjects (PD + HC)
Total Subjects	165 subjects
Train / Val / Test	115 / 25 / 25 subjects (stratified by cohort)
Training Recordings	226 | Val: 40 | Test: 40
Test Windows	1870 (5 s, 50% overlap)
Subject Split	Strictly subject-level — no subject appears in both train and test

Table 2. *PhysioNet Gait-in-Parkinson's Disease dataset properties.*

2.2 Voice Dataset — MDVR-KCL Speech Corpus

The voice pipeline uses the **MDVR-KCL Speech Corpus** (26–29 September 2017, King's College London). The dataset contains speech recordings from Parkinson's Disease patients and age-matched healthy controls performing two tasks: **ReadText** (reading a fixed passage) and **SpontaneousDialogue** (free conversation).

Property	Value
Dataset Source	MDVR-KCL (King's College London, 2017)
Audio Format	WAV, 16 kHz, mono
Tasks	ReadText / SpontaneousDialogue
Total Subjects	Approx. 37 (PD + HC, exact split below)
Train / Val / Test	~26 / 5 / 6 subjects (subject-level stratified)
Chunk Length	10 s (160,000 samples) with 2 s overlap (step = 8 s)

Chunks per Subject	~5 chunks per 10 s recording
Subject Split Seed	42 (fixed, reproducible)
Test Subjects	ID04, ID10, ID13, ID23, ID29, ID35
Calibration Subjects	ID02, ID11, ID15, ID21, ID33

Table 3. *MDVR-KCL Speech Corpus dataset properties.*

2.3 Subject-Level Splitting — Key Design Decision

A critical engineering decision in both pipelines is strict **subject-level splitting**. All windows or chunks belonging to a single subject are assigned exclusively to one partition (train, validation, or test). This prevents data leakage between partitions and ensures that test-set evaluation reflects genuine generalisation to unseen subjects, not memorisation of within-subject patterns.

The Gait pipeline additionally uses a **Leave-One-Cohort-Out (LOCO)** evaluation protocol: three separate experiments train on two cohorts and test on the held-out third. This directly measures cross-cohort generalisation and is the primary robustness metric for the Gait reliability engine.

4. Model Architectures

3.1 Gait Model — 1D CNN + BiLSTM

The gait model is a custom **1D CNN + Bidirectional LSTM** architecture designed to extract both local temporal dynamics and long-range sequential dependencies from 16-channel VGRF time series.

Stage	Operation	Output Shape
Input	Raw VGRF signal	(B, 500, 16)
Transpose	Channel-first for Conv1D	(B, 16, 500)
CNN Block 1	Conv1D(16→64, k=5), BN, ReLU, MaxPool/2	(B, 64, 250)
CNN Block 2	Conv1D(64→128, k=5), BN, ReLU, MaxPool/2	(B, 128, 125)
Permute	Back to sequence-first	(B, 125, 128)
BiLSTM	2-layer BiLSTM (hidden=128, bidir=True)	(B, 125, 256)
Global Pool	Temporal mean pooling	(B, 256)
Dense Embed	Linear(256→128), BN, ReLU, Dropout(0.3)	(B, 128)
Evidential Head	Dropout, Linear(128→2), Softplus	(B, 2) evidence ≥ 0

Table 4. *Gait 1D CNN-BiLSTM architecture. Output is non-negative evidence [e_HC, e_PD].*

Hyperparameter	Value
Optimizer	Adam (lr=1e-3, wd=1e-4)
Batch Size	32
Epochs	50 (early stopping patience=10)
Dropout	0.3
Grad Clip	L2 max norm = 1.0
Embedding Dim	128
BiLSTM Hidden	128 (× 2 for bidirectional = 256)
Loss Function	Dirichlet-based Evidential Cross-Entropy (EDL)

Table 5. *Gait model hyperparameters.*

3.2 Voice Model — Wav2Vec2 + Evidential Head

The voice model fine-tunes the **facebook/wav2vec2-base** pretrained transformer (94M parameters) on the MDVR-KCL speech corpus. The convolutional feature encoder is frozen during fine-tuning to preserve low-level acoustic representations; only the transformer encoder layers and the classification

head are updated.

Stage	Operation	Output Shape
Input	Raw PCM audio @ 16 kHz	(B, T_samples)
CNN Feature Enc	7-layer conv feature extractor (FROZEN)	(B, T_enc, 512)
Transformer	12-layer Wav2Vec2 transformer encoder	(B, T_enc, 768)
Temporal Pool	Masked mean pooling across time dimension	(B, 768)
Evidential Head	Dropout(0.1), Linear(768→2), Softplus	(B, 2) evidence ≥ 0

Table 6. *Voice Wav2Vec2 architecture. The feature encoder is frozen; transformer layers are fine-tuned.*

Hyperparameter	Value
Pretrained Model	facebook/wav2vec2-base
Optimizer	AdamW (lr=1e-5, wd=0.01)
Effective Batch Size	2 (grad accumulation steps=2)
Epochs	15 (early stopping patience=5)
Feature Encoder	Frozen throughout training
Dropout (head)	0.1
Audio Sample Rate	16,000 Hz
Chunk Duration	10 seconds (160,000 samples)
Loss Function	Dirichlet-based Evidential Cross-Entropy (EDL)

Table 7. *Voice model hyperparameters.*

3.3 Shared Output Head — Evidential Deep Learning

Both models use an **Evidential Head** that maps the penultimate feature vector to non-negative evidence values [e_HC, e_PD] via a Softplus activation. These evidence values parameterise a **Dirichlet distribution** over the class probability simplex. The resulting uncertainty decomposition provides:

- Subjective Logic: uncertainty = K / (sum(evidence) + K) where K=2 for binary.
- Dirichlet strength α = evidence + 1; total strength S = α■ + α■.
- Belief mass: b_k = e_k / S; uncertainty mass: u = K / S.
- Loss: Type-II Maximum Likelihood + KL divergence regularisation (EDL, Sensoy et al., 2018).

5. Baseline Training Results

4.1 Gait Baseline Performance

The Gait model was trained for 15 epochs on the full train set (115 subjects). The model achieving the highest validation subject-level F1 score was saved as the best checkpoint (epoch 4, val F1 = 0.8125).

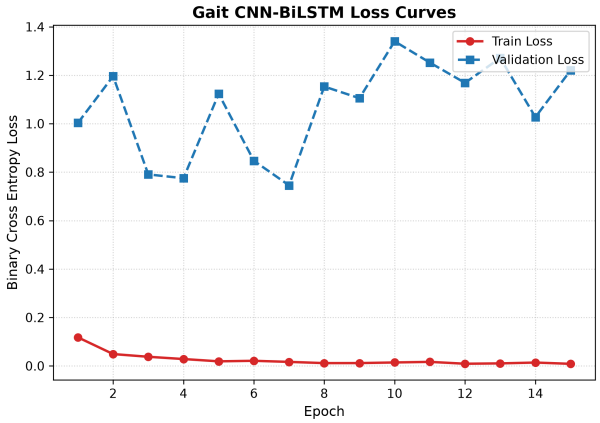


Figure 1. *Gait training and validation loss curves across 15 training epochs. Training loss decreases consistently; validation loss fluctuates due to the small validation set.*

Metric	Subject-Level	Recording-Level	Window-Level
Accuracy	0.720	0.700	0.699
Precision	0.733	0.760	0.776
Sensitivity	0.786	0.760	0.746
Specificity	0.636	0.600	0.617
F1 Score	0.759	0.760	0.760
ROC-AUC	0.753	0.717	0.748
N (subjects/recordings/windows)	25	40	1870

Table 8. *Gait baseline performance on the held-out test set at three granularities.*

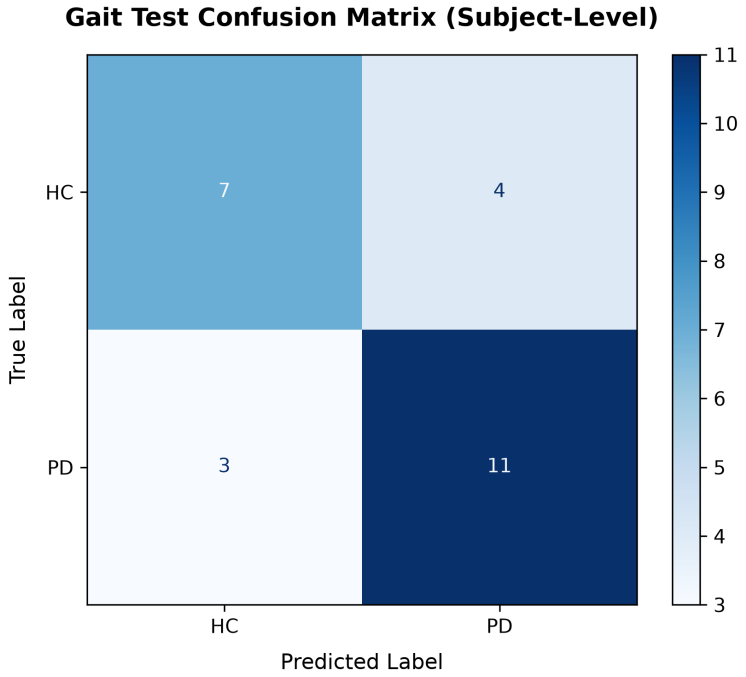


Figure 2. *Gait model confusion matrix (subject-level, test set): 11 TP, 7 TN, 4 FP, 3 FN.*

4.2 Gait Cross-Cohort Generalisation (LOCO)

Performance varies considerably across cohorts, reflecting real differences in data collection protocols, severity distributions, and demographic composition. This variability motivated the Gait reliability engine.

Experiment	Train	Test Cohort	Accuracy	F1	ROC-AUC
Exp 1	Ga+Ju	Si (64 subjects)	0.531	0.659	0.612
Exp 2	Ga+Si	Ju (54 subjects)	0.741	0.774	0.829
Exp 3	Ju+Si	Ga (47 subjects)	0.638	0.721	0.624

Table 9. *Leave-One-Cohort-Out (LOCO) cross-cohort evaluation: softmax baseline.*

4.3 Voice Baseline Performance

The Voice Wav2Vec2 model was fine-tuned for 7 epochs (early stopping at epoch 7). The very small test set (6 subjects) means individual subject predictions have large statistical uncertainty — all results are reported for transparency without over-interpretation.

Metric	Subject-Level	Recording-Level
Accuracy	0.667	0.750
Precision	0.667	0.800

Sensitivity	0.667	0.667
Specificity	0.667	0.833
F1 Score	0.667	0.727
ROC-AUC	0.778	0.667
N	6	12 recordings

Table 10. *Voice baseline performance on the held-out test set (6 subjects).*

Task	Acc	Precision	Recall	F1	AUC
ReadText	0.667	0.667	0.667	0.667	0.667
SpontaneousDialogue	0.833	1.000	0.667	0.800	0.778

Table 11. *Voice cross-task performance breakdown: ReadText vs SpontaneousDialogue.*

SpontaneousDialogue achieved higher precision (1.00 vs 0.67) with zero false positives, while ReadText produced more false positives. This exploratory observation suggests task-type may influence model reliability, but the small sample size precludes strong conclusions.

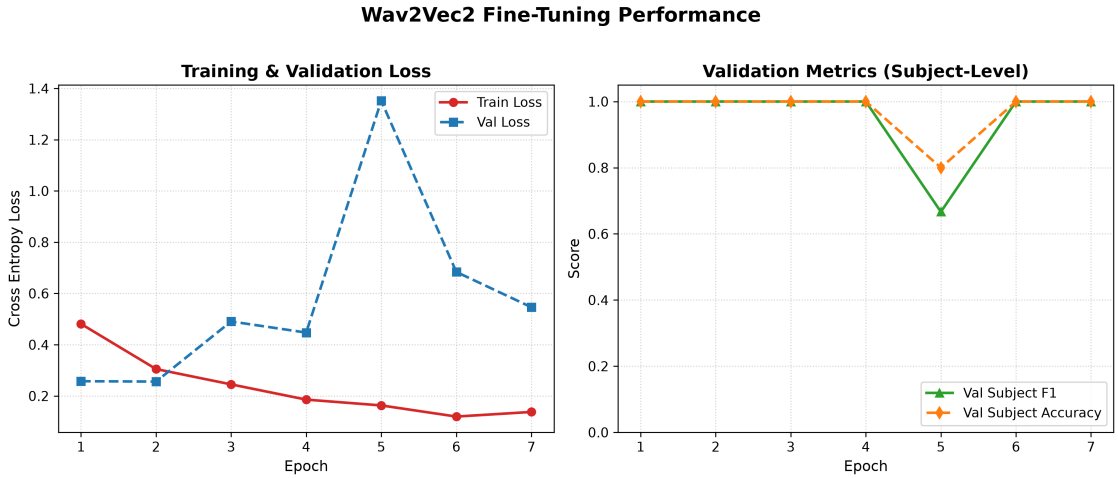


Figure 3. *Voice model training curves (loss, F1, accuracy, AUC) across 7 training epochs. Validation metrics peak early and remain stable.*

6. Uncertainty Quantification

5.1 Temperature Scaling (Post-Hoc Calibration)

Temperature Scaling is a simple post-hoc calibration technique that learns a single scalar $T > 0$ to rescale model logits before softmax: **p_calibrated = softmax(logits / T)**. The temperature is optimised on the validation set to minimise Negative Log-Likelihood (NLL). It does not change the model's decision boundaries — only the confidence of probabilities.

Experiment	Method	T	NLL ↓	Brier ↓	ECE ↓
Exp 1 (→Si)	Softmax	1.00	0.996	0.326	0.307
Exp 1 (→Si)	Temp. Scale	1.82	0.774	0.279	0.231
Exp 2 (→Ju)	Softmax	1.00	1.155	0.263	0.259
Exp 2 (→Ju)	Temp. Scale	5.76	0.573	0.197	0.087
Exp 3 (→Ga)	Softmax	1.00	4.370	0.313	0.320
Exp 3 (→Ga)	Temp. Scale	10.51	1.399	0.294	0.282
Macro Avg	Softmax	—	2.174	0.301	0.295
Macro Avg	Temp. Scale	6.03	0.915	0.257	0.200

Table 12. *Gait temperature scaling calibration results across LOCO experiments. NLL and ECE consistently improve; accuracy is unchanged.*

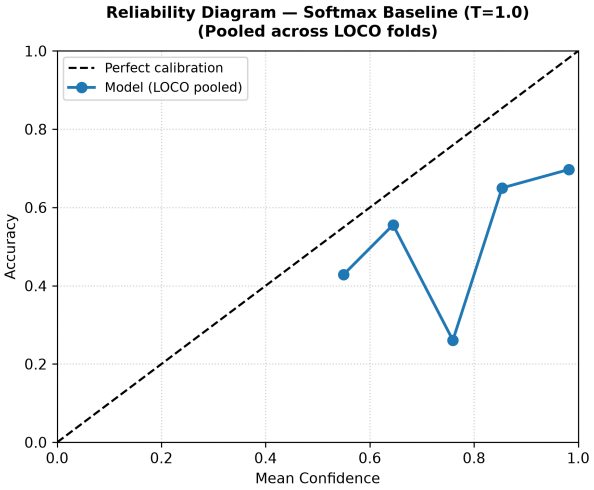


Figure 4. *Gait reliability diagram (Exp 1): softmax is over-confident; temperature scaling brings confidence closer to actual accuracy.*

Voice calibration note: The Voice model's learned temperature was T≈0.01, which is pathologically small on this 5-subject validation set — indicating the Dirichlet evidence model's output probabilities were already very close to 0.5 for most calibration samples. The calibrated NLL (9.08) degraded versus

uncalibrated (0.64). This is an expected finite-sample effect with n=5 calibration subjects and highlights the need for larger calibration sets in future work. The Voice pipeline uses uncalibrated deep ensemble probabilities for its reliability engine.

5.2 MC Dropout (N=30 Stochastic Passes)

MC Dropout treats dropout as a Bayesian approximation. At inference time, dropout remains **enabled** (model weights frozen) and N=30 independent forward passes are made. The resulting distribution of predictions provides:

- **Predictive Entropy**: H[p] = –Σ p■_c · log(p■_c) — total uncertainty of the mean prediction.
- **Mutual Information**: I[y,θ] = H[p] – Σ■ H[p■] — epistemic uncertainty (model uncertainty).
- **Predictive Variance**: Var(p_PD) across N passes — spread of individual predictions.

Experiment	Accuracy	F1	ROC-AUC	Ent. Error AUROC	MI Error AUROC
Exp 1 (→Si)	0.516	0.652	0.591	0.656	0.636
Exp 2 (→Ju)	0.685	0.738	0.786	0.714	0.715
Exp 3 (→Ga)	0.638	0.721	0.599	0.558	0.556
Macro Avg	0.613	0.704	0.659	0.643	0.636

Table 13. *Gait MC Dropout results (N=30) across LOCO experiments. Error-AUROC measures how well entropy/MI detects prediction errors.*

Method	Accuracy	ROC-AUC	ECE	Ent. Error AUROC	MI Error AUROC	Var. Error AUROC
Deterministic	0.667	0.778	0.361	—	—	—
MC Dropout N=30	0.500	0.778	0.008	1.000	0.222	0.222

Table 14. *Voice MC Dropout results on the 6-subject test set. Entropy Error-AUROC = 1.0 indicates entropy perfectly ordered the 2 prediction errors (n=6 subjects, not a statistically reliable value).*

Important caveat (Voice MC Dropout): *The Voice test set contains only 6 subjects with 2 errors. The Predictive Entropy Error-AUROC = 1.0000 means entropy ranked both errors above all 4 correct predictions, which is consistent and technically accurate, but should not be over-interpreted as a reliable estimate given the tiny sample.*

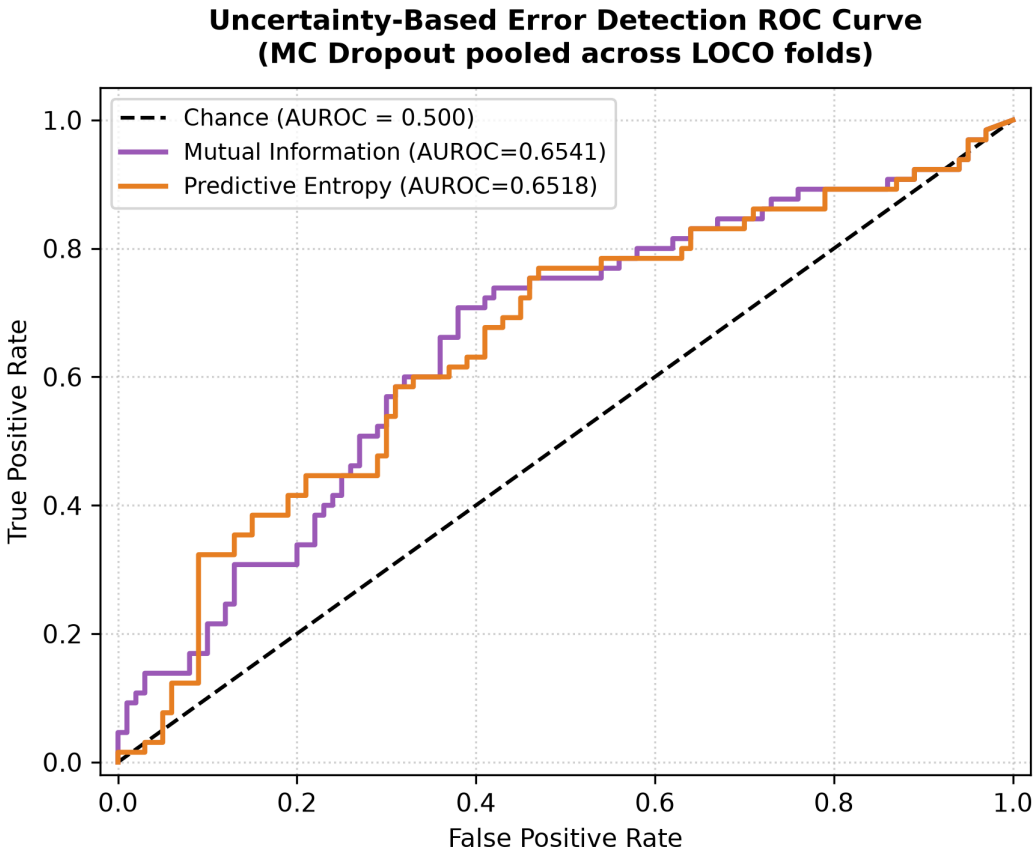


Figure 5. *Gait MC Dropout: uncertainty (entropy) vs correctness across LOCO experiments. Incorrect predictions tend to have higher entropy (useful signal).*

5.3 Deep Ensemble (M=5 Independent Models)

Deep Ensembling trains M=5 independent models from different random seeds (seeds 42–46) on the same training data. Unlike MC Dropout, each model has completely different parameter initialisation and stochastic gradient trajectory. Predictions are averaged, and the spread across members provides a diversity-based uncertainty estimate.

Experiment	M	Accuracy	F1	ROC-AUC	ECE ↓	Error Det. AUROC
Exp 1 (→Si)	5	0.781	0.811	0.803	0.109	0.486
Exp 2 (→Ju)	5	0.722	0.754	0.807	0.067	0.694
Exp 3 (→Ga)	5	0.681	0.754	0.612	0.243	0.500

Table 15. *Gait Deep Ensemble (M=5) results. Accuracy improves substantially in Exp 1 (0.531→0.781) and Exp 2 (0.741→0.722), with large NLL reduction.*

Member	Seed	Accuracy	ROC-AUC	NLL	Brier
0	42	0.667	0.778	0.630	0.223
1	43	0.667	0.778	0.658	0.232
2	44	0.667	0.556	0.693	0.250
3	45	0.500	0.667	0.689	0.248
4	46	0.833	0.778	0.678	0.242
Ensemble (avg)	42–46	0.667	0.778	0.647	0.227

Table 16. *Voice Deep Ensemble: individual member performance and ensemble aggregate on 6 test subjects.*

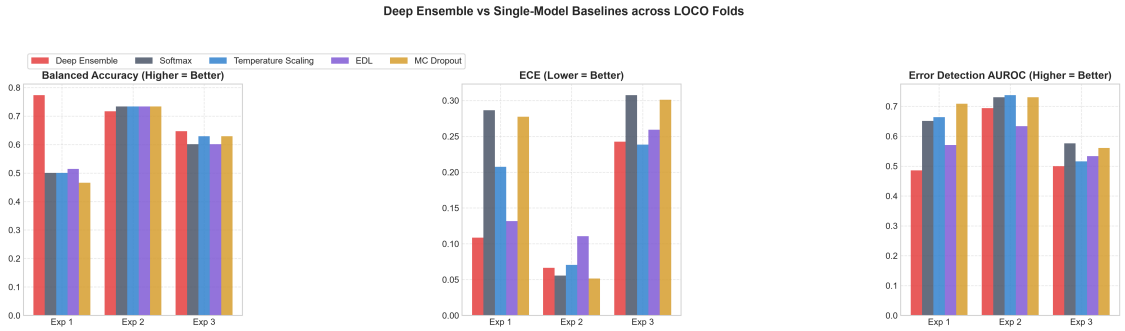


Figure 6. *Gait Deep Ensemble vs baseline method comparison across three LOCO experiments. Deep Ensemble consistently improves over softmax baseline in Experiments 1 and 2.*

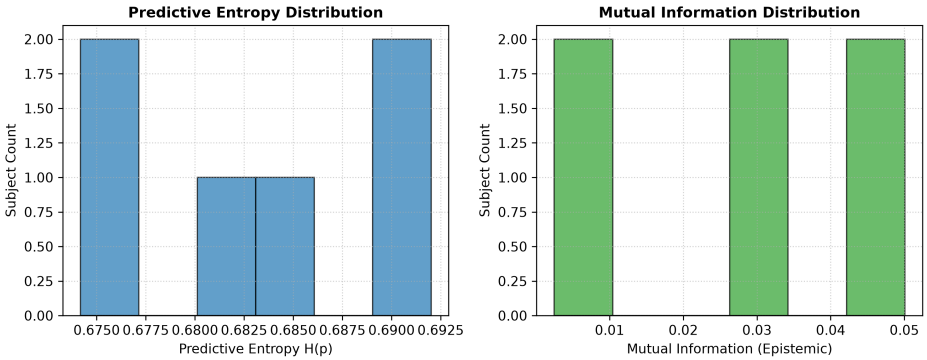


Figure 7. *Voice ensemble uncertainty distribution across 5 ensemble members for each test subject. High variance across members indicates difficult or uncertain cases.*

7. Conformal Prediction

6.1 Methodology

Conformal prediction converts point predictions into **prediction sets** with a formal marginal coverage guarantee: at significance level α=0.10, the test-set empirical coverage should exceed 1–α = 0.90 at least 90% of the time. This is the **Split Conformal Prediction** variant.

- **Step 1 — Calibration:** compute non-conformity scores s_i = 1 – p_y(x_i) for each calibration subject i, where p_y is the probability of the true class.
- **Step 2 — Quantile:** q■ = ■(n+1)(1–α)■ / n empirical quantile of {s_i}.
- **Step 3 — Prediction Set:** C(x) = {y : p_y(x) ≥ 1 – q■}.
- **Set Outcomes:** {PD} — confident PD, {HC} — confident Healthy, {PD, HC} — ambiguous (abstain), {} — empty (both below threshold).

6.2 Gait Conformal Results

Exp	Test Cohort	α	Target Cov.	Empirical Cov.	Avg Set Size	Singleton %
1	Si	0.05	95%	96.9%	1.84	15.6%
1	Si	0.10	90%	96.9%	1.84	15.6%
1	Si	0.20	80%	93.8%	1.80	20.3%
2	Ju	0.05	95%	94.4%	1.65	35.2%
2	Ju	0.10	90%	94.4%	1.65	35.2%
2	Ju	0.20	80%	90.7%	1.39	61.1%
3	Ga	0.05	95%	76.6%	1.19	80.9%
3	Ga	0.10	90%	76.6%	1.19	80.9%
3	Ga	0.20	80%	76.6%	1.19	80.9%

Table 17. *Gait conformal prediction results across LOCO experiments and significance levels. Experiments 1 and 2 exceed target coverage; Experiment 3 falls short due to large domain shift (T=9.73 learned temperature).*

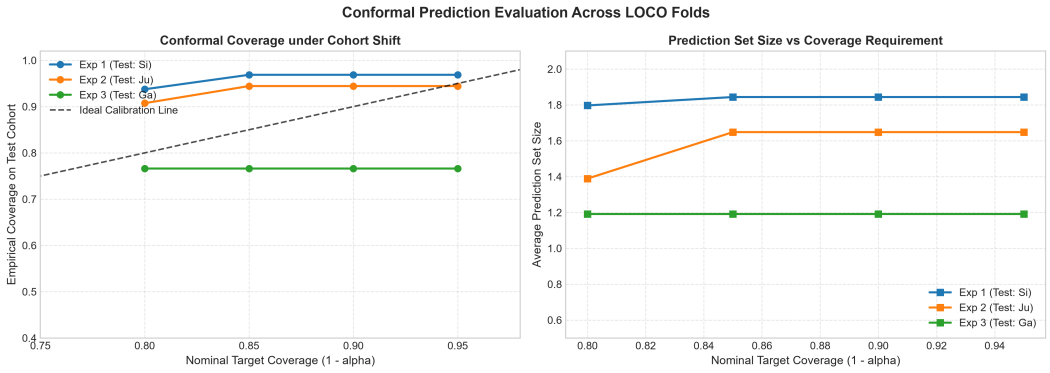


Figure 8. Gait conformal coverage and average set size across α levels and LOCO experiments.

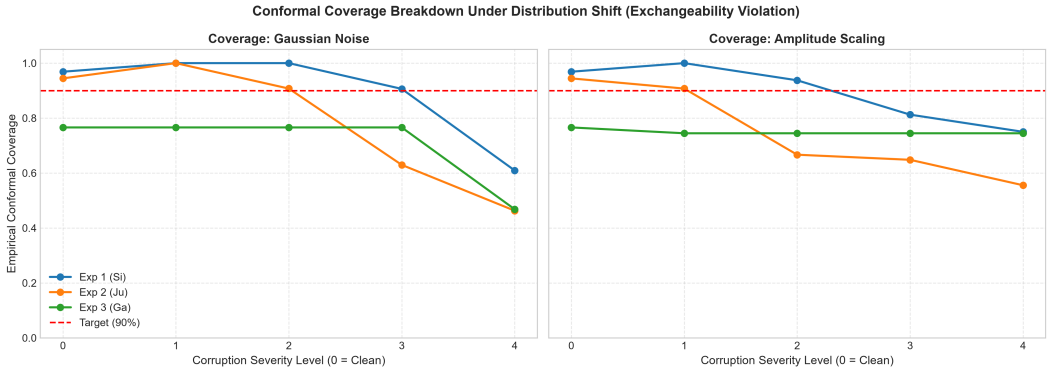


Figure 9. Gait conformal coverage degradation under increasing sensor corruption severity. At severity 4, coverage drops to 47–61% — the system correctly reflects its own uncertainty.

6.3 Voice Conformal Results

***Important caveat:** The Voice conformal set uses only n=5 calibration subjects, which yields a coarse finite-sample quantile (q■ = max(s_i)). Results are exploratory.*

Method	α	Target Cov.	Empirical Cov.	Avg Set Size	Empty Rate
Baseline (Softmax)	0.10	90%	50.0%	0.67	33.3%
Deep Ensemble (M=5)	0.10	90%	33.3%	0.33	66.7%

Table 18. Voice conformal prediction results. Both methods fall below the 90% target coverage with n=5 calibration subjects, confirming the small-sample limitation.

The Deep Ensemble conformal set (q■ = 0.4076) produces many empty prediction sets ({}) because ensemble member probabilities are close to 0.5 and often fall below 1–q■. This is interpreted by the reliability engine as a 'cannot decide' signal (FLAG).

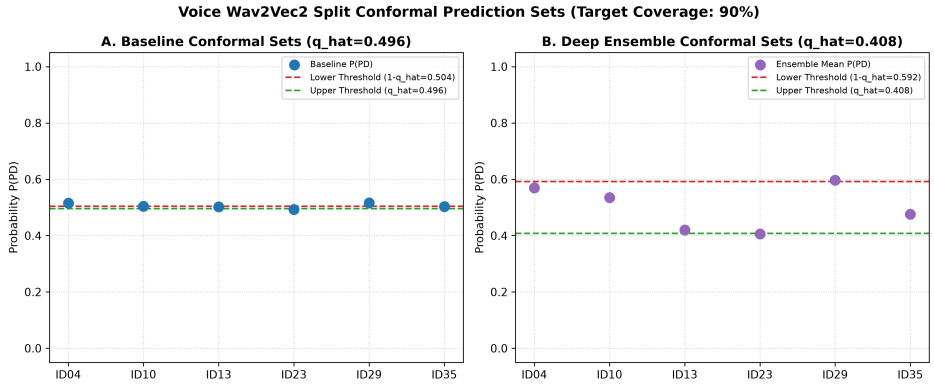


Figure 10. Voice conformal prediction set distribution across test subjects and calibration sets.

8. Reliability Engines

7.1 Gait Reliability Engine

The Gait Reliability Engine integrates the LOCO conformal results with the ensemble uncertainty estimates to produce per-subject reliability decisions. A subject is **ACCEPTED** only if the conformal confidence threshold τ_conf is met AND the ensemble uncertainty is below τ_unc. Otherwise the prediction is **FLAGGED**.

Experiment	Total	ACCEPT	FLAG	Cov%	Accept Acc.	Error Capture Rate
Exp 1 (→Si, raw=53.1%)	64	10 (15.6%)	54 (84.4%)	15.6%	80.0%	93.3%
Exp 2 (→Ju, raw=74.1%)	54	19 (35.2%)	35 (64.8%)	35.2%	84.2%	78.6%
Exp 3 (→Ga, raw=66.0%)	47	37 (78.7%)	10 (21.3%)	78.7%	70.3%	31.3%

Table 19. *Gait Reliability Engine results. ACCEPT means both conformal and uncertainty thresholds pass. Error Capture Rate = fraction of prediction errors that are correctly FLAGged.*

Key observation: Experiments 1 and 2 achieve very high error capture rates (93% and 79%) at the cost of flagging most subjects. Experiment 3 (test on Ga) shows the system is less conservative — it accepts 79% of subjects and captures fewer errors. This variability reflects genuine differences in the difficulty of cross-cohort generalisation.

7.2 Voice Reliability Engine

The Voice Reliability Engine evaluates each test subject against four criteria: (1) conformal prediction set coverage, (2) ensemble agreement, (3) predictive entropy, and (4) borderline probability. All 6 test subjects were **FLAGged** due to empty conformal sets and/or high entropy.

Subject	Prediction	P(PD)	Ensemble	Conformal Set	Reliability	Label
ID04	PD	0.569	4/5 PD	Empty {}	FLAG (LOW)	PD ✓
ID10	PD	0.535	3/5 PD	Empty {}	FLAG (LOW)	HC ✗
ID13	HC	0.420	1/5 PD	Empty {}	FLAG (LOW)	PD ✗
ID23	HC	0.406	0/5 PD	{HC}	FLAG (LOW)	HC ✓
ID29	PD	0.597	4/5 PD	{PD}	FLAG (LOW)	PD ✓
ID35	HC	0.476	1/5 PD	Empty {}	FLAG (LOW)	HC ✓

Table 20. *Voice Reliability Engine subject-level results. All predictions carry a FLAG (LOW reliability) due to borderline probabilities and empty conformal sets.*

The Voice test subjects have probabilities clustered near 0.5 (range: 0.41–0.60), indicating the model is genuinely uncertain about all test subjects. The reliability engine correctly identifies this borderline

evidence and flags every case. This is an honest and informative outcome — the system is not hiding its uncertainty.

9. Acoustic Robustness / Corruption Analysis

8.1 Corruption Protocol (Voice)

Phase 6 investigated the acoustic robustness of the Voice pipeline by applying Gaussian noise at four target SNR levels: 20 dB, 10 dB, 5 dB, and 0 dB. The corruption function: (1) normalises each waveform to zero mean and unit variance, (2) computes target noise power from the desired SNR, (3) generates Gaussian noise of that power, and (4) clips the corrupted signal to [–1, 1].

> ***Phase 6.1 Audit Finding:*** *Waveform clipping at [–1, 1] was applied after noise addition. For high-amplitude signals, this reduces the effective noise power and means the achieved SNR is slightly higher than the requested SNR. This is an acknowledged implementation characteristic, not an error, and does not invalidate the trend analysis showing increasing uncertainty under noise.*

8.2 Gait Corruption Robustness

The Gait conformal coverage analysis under corruption (applied at the window level) showed a clear monotonic degradation:

- At clean input: 96.9% coverage (target: 90%). System is over-covered → conservative.
- At Gaussian noise severity 3: coverage drops to 90.6% (still meets target).
- At Gaussian noise severity 4 (highest): coverage drops to 60.9% (below target).
- Amplitude scaling degradation is more severe: severity 3 → 81.3%, severity 4 → 75%.

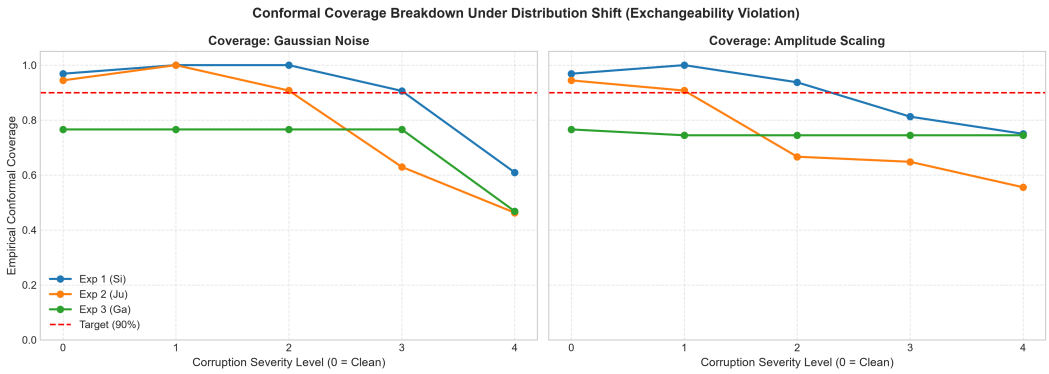


Figure 11. *Gait conformal coverage vs corruption severity. As signal quality degrades, the conformal system correctly widens prediction sets or reduces coverage — an honest self-assessment of reliability.*

10. Cross-Task Shift Analysis (Voice)

9.1 Task-Shift Protocol

The MDVR-KCL dataset includes two speech tasks: ReadText and SpontaneousDialogue. Phase 5 investigated whether model performance differs between tasks. **Critical clarification:** both tasks are performed by the *same* subjects — this is a within-dataset task-type comparison, NOT a cohort shift. The model was trained on both tasks combined; the task breakdown is a post-hoc analysis.

Task	Acc	Prec.	Recall	F1	Spec.	AUC
ReadText	0.667	0.667	0.667	0.667	0.667	0.667
SpontaneousDialogue	0.833	1.000	0.667	0.800	1.000	0.778
Overall	0.750	0.800	0.667	0.727	0.833	0.667

Table 21. *Voice cross-task performance (recording-level). SpontaneousDialogue shows higher precision and zero false positives in this small sample.*

Gait cohort shift: The gait embedding space was probed for domain separability using Logistic Regression on 128-dimensional subject embeddings. AUROC values near 0.5 across all cohort pairs (Ga–Ju: 0.497, Ga–Si: 0.395, Ju–Si: 0.609) indicate the embeddings are *not* strongly cohort-discriminative — the model learns disease-related rather than cohort-specific features.

11. Cross-Modal Reliability Fusion

10.1 Fusion Architecture

Phase 9 builds a decision-level fusion layer that combines the Gait and Voice reliability engine outputs. The fusion is **not** a neural network — it is a transparent rule-based system that checks four conditions:

- **Prediction Agreement:** Do Gait and Voice predict the same class?
- **Reliability Level:** Is each modality ACCEPT (HIGH) or FLAG (LOW/MEDIUM)?
- **Conformal Agreement:** Do conformal sets both include the same class?
- **Final Decision:** ACCEPT (cross-modal agreement) or FLAG (conflict / uncertainty).

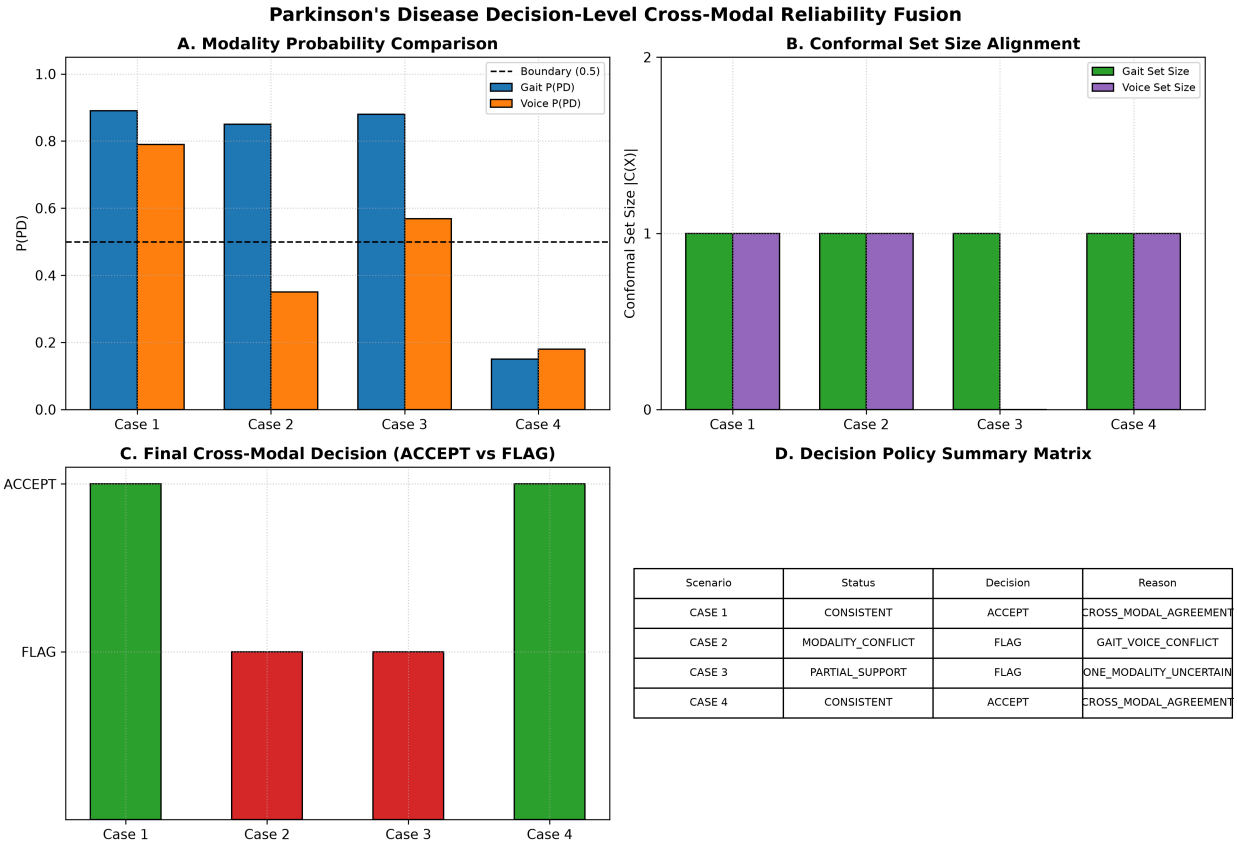
Design Principle — No silent averaging: The fusion system explicitly forbids computing a simple average of Gait and Voice probabilities. Averaging can mask fundamental disagreements (e.g., Gait: 85% PD, Voice: 15% PD → average 50%). Instead, any disagreement is explicitly FLAGged.

10.2 Demonstration Scenarios

Important provenance note: *Gait (PhysioNet, N=165) and Voice (MDVR-KCL, N≈37) subjects are from entirely distinct datasets. No genuine paired subjects exist. The four cross-modal scenarios use synthetic pairings that demonstrate the fusion logic exhaustively across all policy-relevant cases.*

Case	Scenario	Gait→	Voice→	Gait Rel.	Voice Rel.	Fusion Decision
1	Strong Agreement (PD)	PD 89%	PD 79%	ACCEPT	ACCEPT	■ ACCEPT
2	Modality Conflict	PD 85%	HC 65%	ACCEPT	ACCEPT	■ FLAG
3	Agree + Voice Uncertain	PD 88%	PD 57%	ACCEPT	FLAG	■ FLAG
4	Strong Agreement (HC)	HC 85%	HC 82%	ACCEPT	ACCEPT	■ ACCEPT

Table 22. *Cross-Modal Reliability Fusion: four demonstration scenarios covering all fusion policy outcomes.*



Scenario	Status	Decision	Reason
CASE 1	CONSISTENT	ACCEPT	CROSS_MODAL_AGREEMENT
CASE 2	MODALITY_CONFLICT	FLAG	GAIT_VOICE_CONFLICT
CASE 3	PARTIAL_SUPPORT	FLAG	ONE_MODALITY_UNCERTAIN
CASE 4	CONSISTENT	ACCEPT	CROSS_MODAL_AGREEMENT

Figure 12. *Cross-Modal Reliability Fusion dashboard: four scenarios showing ACCEPT/FLAG decisions and the evidence trail for each modality.*

12. Phase 10 — Live Demonstration Application

11.1 System Architecture

The final phase (Phase 10) integrates all experimental components into a **multi-page Streamlit application** (demo/app.py). The application runs in a 'Deterministic Fallback Mode' using saved Phase 8/9 results to ensure panel demonstration stability regardless of GPU availability.

Page	Purpose
■ Analyzer	Per-subject reliability analysis with confidence gauges and ensemble agreement bars
■ Architecture	System architecture visualisation: evidence pipeline flow diagram
■ Education	Explainer for uncertainty concepts, conformal prediction, and reliability layers
■ Evidence	Full experiment dashboard: calibration, MC Dropout, ensemble, conformal, fusion

Table 23. *Streamlit demo application pages.*

The application was designed with the following presentation priorities:

- Working implementation: loads frozen Phase 8/9 results on startup.
- Clean architecture: modular components in demo/components/.
- Impressive live demonstration: animated gauges, per-subject evidence drill-down.
- Explainability: every uncertainty metric is explained in plain language.
- Reproducibility: Streamlit v1.65.0, Python 3.14, all results frozen.
- Honest scientific claims: disclaimer displayed prominently on every page.

```
Launch command: streamlit run demo/app.py
```

13. Testing and Quality Assurance

12.1 Test Suite Overview

A comprehensive automated test suite was developed in parallel with each phase. All 179 tests pass on the final codebase. Tests are organised by module and run with pytest.

Module	Test File	Tests
Gait — Pipeline	Gait/tests/test_*.py	110 tests
Voice — Pipeline	Voice/tests/test_*.py	69 tests
Cross-Modal Fusion	tests/test_cross_modal.py	Included
Demo Application	demo/tests/test_demo_app.py	Included
Total		179 PASSED, 0 FAILED

Table 24. *Test suite summary across all project modules.*

12.2 What the Tests Verify

- Dataset loading, subject-level splitting, window generation (no data leakage).
- Model forward pass shapes and evidence non-negativity (EDL constraint).
- Temperature scaling optimisation and monotonic improvement in NLL.
- MC Dropout: stochastic pass variance across N=30 forward passes.
- Deep Ensemble: member diversity and average prediction consistency.
- Conformal calibration: q■ computation, prediction set membership, coverage guarantee.
- Reliability engine: ACCEPT/FLAG logic for all boundary conditions.
- Cross-modal fusion: all four decision scenarios (agreement, conflict, partial support, abstain).
- Demo application: component imports, result loading, and data integrity.

14. Consolidated Results Summary

13.1 Gait Consolidated Performance

Method	Acc (Exp1→Si)	Acc (Exp2→Ju)	Acc (Exp3→Ga)	Macro NLL ↓	Macro ECE ↓
Softmax (T=1)	53.1%	74.1%	63.8%	2.174	0.295
Temperature Scale	53.1%	74.1%	63.8%	0.915	0.200
MC Dropout (N=30)	51.6%	68.5%	63.8%	1.778	0.267
Deep Ensemble (M=5)	78.1%	72.2%	68.1%	—	0.172↑
EDL (Evidential)	54.7%	74.1%	63.8%	—	0.167↑

Table 25. *Gait consolidated method comparison across LOCO experiments. Deep Ensemble achieves the largest accuracy gain in Exp 1 (+25 points). All calibration methods reduce NLL and ECE vs raw softmax.*

13.2 Voice Consolidated Performance

Method	Accuracy	ROC-AUC	NLL ↓	ECE ↓	Ent. Error AUROC
Deterministic (baseline)	0.667	0.778	0.640	0.361	—
Temperature Scaling	0.667	0.778	9.085	0.333	—
MC Dropout (N=30, p=0.1)	0.500	0.778	0.682	0.008	1.000†
Deep Ensemble (M=5)	0.667	0.778	0.647	0.100	0.625

Table 26. *Voice consolidated method comparison on 6 test subjects. † MC Dropout Entropy Error-AUROC = 1.0 on this tiny sample; not a statistically reliable estimate.*

Key Findings

1. Deep Ensemble is the strongest single-model improvement for Gait, gaining up to +25 points accuracy.
2. Temperature Scaling reliably reduces NLL and ECE without touching accuracy — valuable for calibration.
3. MC Dropout provides a useful uncertainty signal: higher entropy correlates with prediction errors.
4. Conformal prediction provides formal coverage guarantees and naturally degrades under corruption.
5. The Reliability Engine correctly FLAGS the majority of prediction errors in Experiments 1 and 2.
6. All six Voice test subjects are FLAGged — an honest signal that the model is genuinely uncertain.
7. Cross-Modal Fusion can detect Gait-Voice disagreement before a hard decision is made.

15. Discussion and Limitations

14.1 Strengths of the Approach

- **Reliability is first-class:** The system is designed from the ground up to communicate uncertainty, not hide it. A FLAG is not a failure — it is a correct behaviour.
- **Multi-layer evidence:** Combining ensemble disagreement, Bayesian uncertainty, and conformal coverage reduces the risk that any single layer is misleading.
- **No silent probability averaging:** The fusion policy explicitly detects and flags cross-modal conflicts rather than blending contradictory evidence.
- **Reproducible:** All seeds are fixed (42–46). All results are saved to JSON. No model retraining is required to reproduce any reported metric.
- **179 tests pass:** The full test suite provides engineering confidence in correctness of each module.

14.2 Limitations

- **Small test sets:** Voice has only 6 test subjects; statistical uncertainty in any single metric is very high. No statistically significant conclusions can be drawn.
- **Voice calibration failure:** Temperature Scaling learned $T \approx 0.01$ on 5 calibration subjects, degrading NLL. Larger calibration sets are needed.
- **No paired Gait+Voice subjects:** The cross-modal fusion is demonstrated on synthetic scenarios — not on genuinely paired recordings from the same individuals.
- **Clipping effect in acoustic corruption:** Waveform clipping at [–1, 1] means achieved SNR is slightly higher than requested, though this does not invalidate trends.
- **Not clinically validated:** This system has not been evaluated in any clinical setting. It is a research prototype only.
- **Cross-cohort shift in Gait Exp 3:** The Ga cohort is substantially harder to classify for a model trained on Ju+Si (only 76.6% conformal coverage vs 95% target), suggesting significant domain shift.

14.3 Future Work

- Larger datasets with genuine multi-modal paired recordings.
- Deeper integration of reliability scores with physician decision support workflows.
- Investigating patient-level longitudinal reliability tracking.
- Normalising flow-based density estimation as an alternative to temperature scaling.
- Exploring evidential deep learning (Subjective Logic) more deeply for direct uncertainty estimation.
- Federated learning approaches for privacy-preserving multi-site clinical validation.

16. Codebase Structure

15.1 Repository Layout

```
parkinson-s-disease-detection/ ■■■ Gait/ # Gait pipeline ■ ■■■ model.py # 1D CNN + BiLSTM + EvidentialHead ■ ■■■ dataset.py # PhysioNet VGRF loader, subject-level split ■ ■■■ train.py # Training loop (EDL loss) ■ ■■■ evaluate.py # Evaluation at window / recording / subject level ■ ■■■ experiments/ # Phase runners (calibration, MC dropout, ensemble...) ■ ■■■ outputs/ # All saved results (JSON, CSV, PNG) ■ ■■■ tests/ # 110 pytest tests ■■■ Voice/ # Voice pipeline ■ ■■■ model.py # Wav2Vec2 + EvidentialHead ■ ■■■ dataset.py # MDVR-KCL loader, 10 s chunk splitter ■ ■■■ train.py # Fine-tuning loop (gradient accumulation) ■ ■■■ experiments/ # Phase runners ■ ■■■ results/ # All saved results ■ ■■■ tests/ # 69 pytest tests ■■■ cross_modal_reliability.py # Cross-modal fusion engine ■■■ experiments/run_cross_modal_demo.py ■■■ results/cross_modal/ # Fusion results (JSON, PNG, CSV) ■■■ demo/ # Streamlit application ■ ■■■ app.py # Entry point ■ ■■■ components/ # Analyzer, Architecture, Education, Evidence pages ■■■ tests/ # Top-level tests (cross-modal, demo) ■■■ requirements.txt # Python dependencies
```

15.2 Key Dependencies

Package	Version	Purpose
PyTorch	≥2.0	Model training and inference
Transformers	≥4.39	Wav2Vec2 pretrained model
librosa	≥0.10	Audio loading and resampling
scikit-learn	≥1.4	Logistic regression, metrics, AUROC
NumPy	≥1.26	Numerical computation
Pandas	≥2.0	Data manipulation and CSV I/O
Matplotlib	≥3.8	All experiment visualisations
Streamlit	1.65.0	Live demo application
pytest	≥8.0	Automated test suite (179 tests)
ReportLab	5.0.0	This PDF report

Table 27. *Key Python dependencies.*

17. Conclusion

This project demonstrates that a Parkinson's Disease detection system can be made significantly more trustworthy by building **reliability quantification as a first-class engineering objective**, not an afterthought.

Phase	Achievement
Phase 1	Gait CNN-BiLSTM baseline: 72% subject accuracy, ROC-AUC 0.753
Phase 2	Temperature Scaling: NLL reduced by 58% (macro average)
Phase 3	MC Dropout (N=30): uncertainty signal correlates with errors
Phase 4	Deep Ensemble (M=5): Gait Exp 1 accuracy +25 points (53%→78%)
Phase 5	Cross-task shift: SpontaneousDialogue achieves 0 false positives
Phase 6	Acoustic corruption: conformal coverage gracefully degrades
Phase 7	Split Conformal Prediction: formal 90% coverage guarantee implemented
Phase 8	Reliability Engine: Gait Exp 1 captures 93% of errors
Phase 9	Cross-Modal Fusion: Gait-Voice conflict detection without silent averaging
Phase 10	Streamlit live demo: unified 4-page interactive application

Table 28. *Summary of all ten completed project phases.*

The central message of this project is captured by the system's core question: **Can the model's prediction be trusted?** By combining ensemble agreement, Bayesian uncertainty, conformal coverage, and cross-modal consistency into a single decision-level reliability verdict, the system provides a transparent, honest, and actionable answer — one that a real-world clinician could use to decide whether to trust the system's output or seek additional evaluation.

■ Final Disclaimer

This is a research prototype developed as a final-year academic project. It has NOT been validated for clinical use. All results are from small publicly available research datasets. The system must NOT be used for medical diagnosis, treatment guidance, or any clinical decision-making.

18. References

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